



Student Number:

Teacher:

St George Girls High School

# Mathematics Advanced

2024

Trial HSC Examination

## General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- For questions in **Section I**, use the Multiple-Choice answer sheet provided at the back of this booklet.

For questions in **Section II**:

- Answer the question in the spaces provided.
- **Show relevant mathematical reasoning** and/or calculations.
- **Extra writing space** is provided at the back of each booklet. If you use this space, clearly indicate which question you are answering.
- Marks may not be awarded for incomplete or poorly presented solutions or where multiple solutions are provided.

**Total marks:**  
**100**

### Section I – 10 marks (pages 2 to 7)

- Attempt Questions 1-10
- Allow about 20 minutes for this section.

### Section II – 90 marks (pages 9 to 40)

- Attempt Questions 11 – 35
- Allow about 2 hours and 40 minutes for this section.

|              |             |
|--------------|-------------|
| Q1 – Q10     | /10         |
| Q11 – Q17    | /20         |
| Q18 – Q23    | /18         |
| Q24– Q28     | /17         |
| Q29 – Q32    | /17         |
| Q33 – Q35    | /18         |
| <b>Total</b> | <b>/100</b> |
|              | %           |

## **Section I**

### **10 marks**

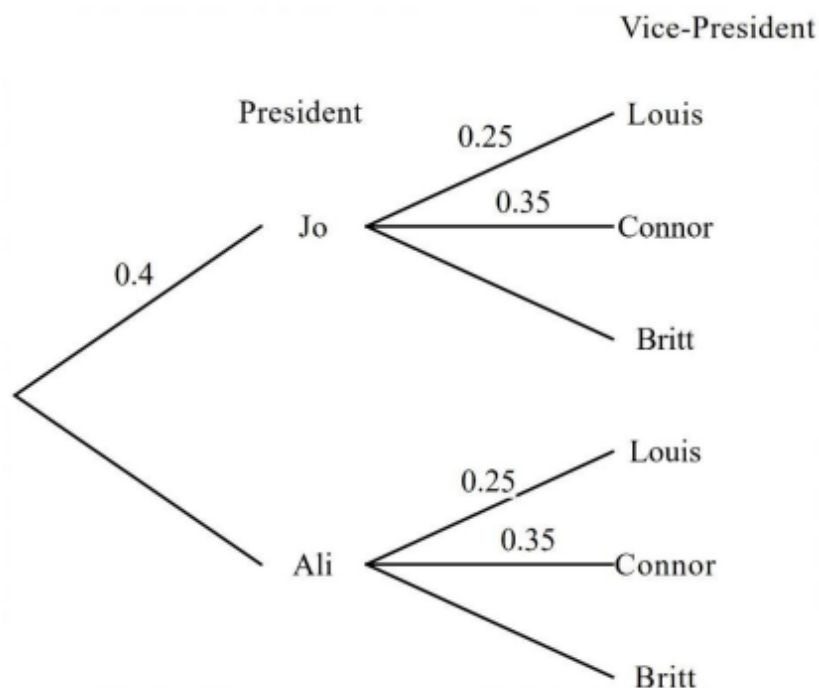
Attempt Questions 1 – 10.

Allow about **20 minutes** for this section.

Use the **multiple-choice answer sheet** for questions 1 – 10.

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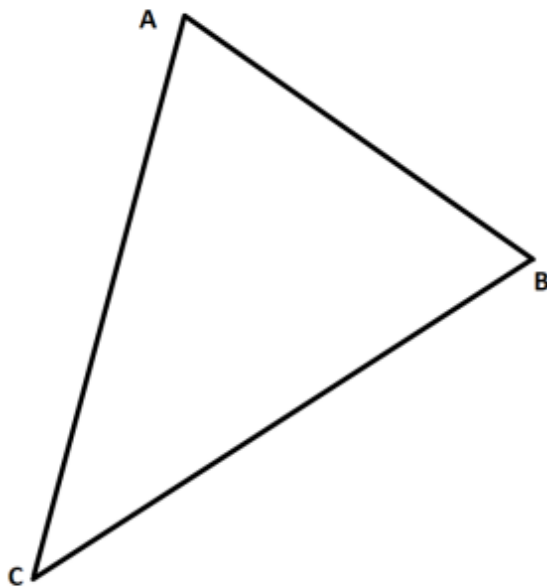
1. The partially completed probability tree below shows the results of a survey of voting intentions of shareholders in an election for President and Vice-President of a large corporation.



Based on this survey, what is the probability that Jo is elected President, with either Britt or Louis as Vice-President?

- (A) 0.04
- (B) 0.26
- (C) 0.30
- (D) 0.50

2. Which one of the following statements about the line  $12x - 4y = 0$  is **NOT** true?
- (A) The line has a slope of 12
- (B) The line passes through the origin
- (C) The line has the same slope as the line with the equation  $12x - 4y = 12$
- (D) For this line, as  $x$  increases  $y$  increases
3. On her bushwalk, Angela walks 2.5 km from A to B on a bearing of  $140^\circ$  and then walks from B to C for 3.5 km on a bearing of  $235^\circ$ .



How far, to one decimal place, does Angela need to walk to return to the starting point?

- (A) 5.2 km
- (B) 4.4 km
- (C) 4.1 km
- (D) 3.1 km

4. The parabola  $f(x) = -x^2 + 9mx + 2$  has a maximum at  $x = \frac{2}{m}$ , where  $m$  is a constant such that  $m \neq 0$ .

What are the values of  $m$ ?

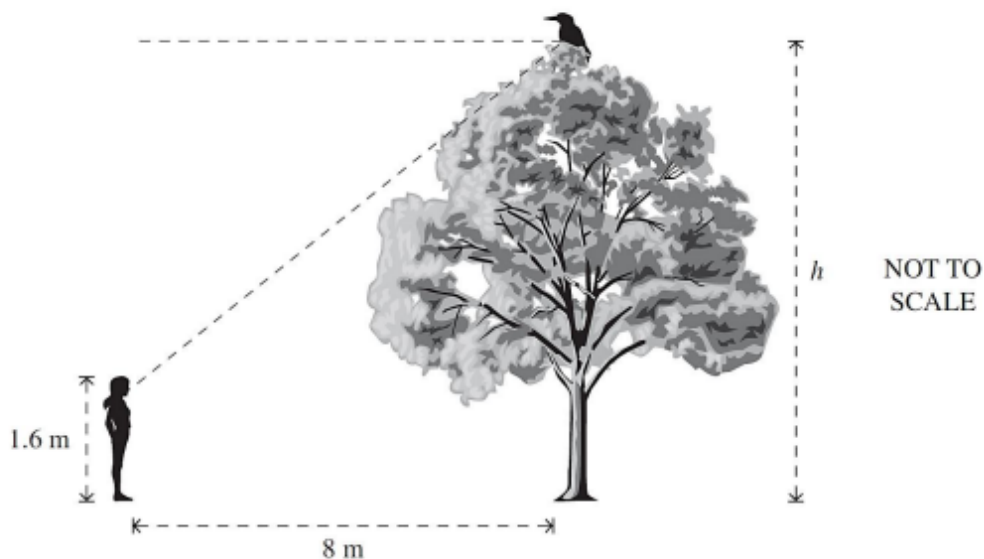
- (A)  $m = \frac{2}{3}$  or  $m = -\frac{2}{3}$
- (B)  $m = \frac{2}{9}$  or  $m = -\frac{2}{9}$
- (C)  $m = \frac{9}{4}$  or  $m = -\frac{9}{4}$
- (D)  $m = \frac{3}{2}$  or  $m = -\frac{3}{2}$
5. What are the coordinates of the turning point to the curve  $y = e^x - ex$ ?
- (A)  $(0, 1)$
- (B)  $(e, 1)$
- (C)  $(1, e)$
- (D)  $(1, 0)$

6. A series has following terms:

$$2+5+8+11+14+17+ \dots$$

What is the sum of the first 30 terms of the series

- (A) 1410  
(B) 1365  
(C) 1380  
(D) 2820
7. Kelly is watching a bird at the top of a tree. Kelly is 1.6 m tall and is standing 8 m away from the tree, as shown in the diagram.



If the angle of depression of Kelly from the bird is  $73^\circ$ , what is the height of the tree?

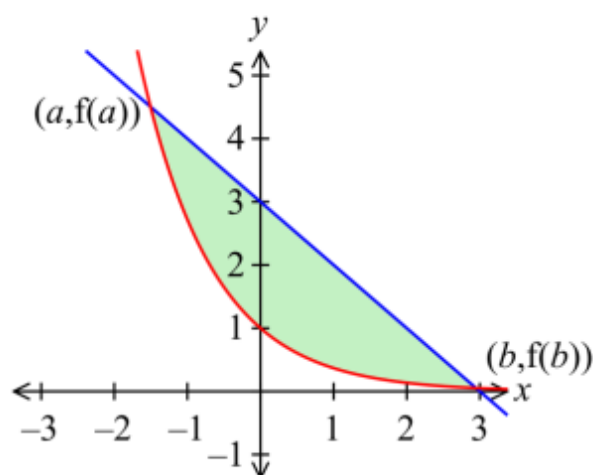
- (A) 2.45 m  
(B) 4.05 m  
(C) 26.17 m  
(D) 27.77 m

8. The functions  $g(x)$  and  $h(x)$  are defined as follows:

$$g(x) = \frac{x^2}{8} \qquad h(x) = \sqrt{2x} + 3$$

Which is the correct expression for  $h(g(x))$ ?

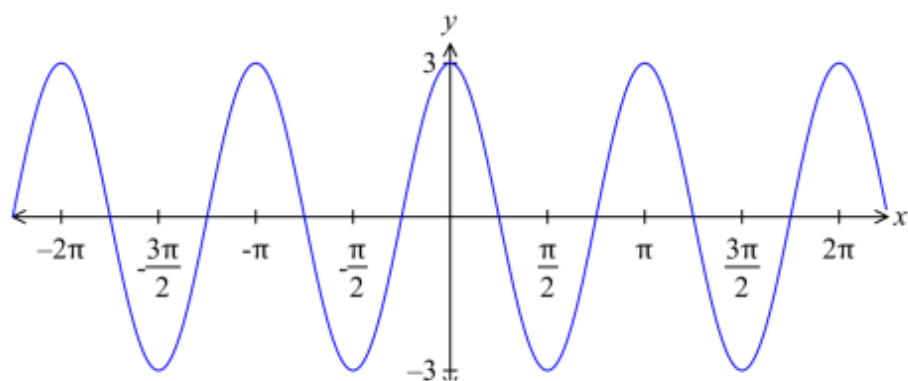
- (A)  $\sqrt{\frac{2x}{3}}$
- (B)  $\frac{x}{2} - 3$
- (C)  $\frac{x}{2} + 3$
- (D)  $\sqrt{\frac{x}{2}} + 3$
9. The diagram below shows the region enclosed by  $x + y = 3$  and  $y = e^{-x}$ .



Which of the following integrals represents the area of the shaded region?

- (A)  $\int_a^b (e^{-x} - 3 + x) dx$
- (B)  $\int_a^b (3 - x + e^{-x}) dx$
- (C)  $\int_a^b (e^{-x} + 3 + x) dx$
- (D)  $\int_a^b (3 - x - e^{-x}) dx$

10. The graph of  $y = f(x)$  is show below.



Which of the following could **NOT** be the correct description of  $f(x)$  ?

- (A)  $y = 3\sin 2(x + \frac{\pi}{2})$
- (B)  $y = 3\sin(2x + \frac{\pi}{2})$
- (C)  $y = 3\cos 2x$
- (D)  $y = 3\cos 2(x - \pi)$

**END OF SECTION I**

# Mathematics Advanced

## Section II Answer Booklet 1

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| Student Number: |
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### Section II

**90 marks**

**Attempt Questions 11 – 35**

**Allow about 2 hours 40 minutes for this section**

**Booklet 1 – Attempt Question 11 – 28 (55 marks)**

**Booklet 2 – Attempt Question 29 – 35 (35 marks)**

#### Instructions

- Write your Teacher's Name and Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided on pages 26 & 27 of Booklet 1. If you use this space, clearly indicate which question you are answering.

|           |     |
|-----------|-----|
| Q11 – Q17 | /20 |
| Q18 – Q23 | /18 |
| Q24 – Q28 | /17 |

**Please turn over**

## **Section II**

**90 marks**

**Attempt questions 11 - 35**

**Allow about 2 hours and 40 minutes for this section**

Answer each question in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

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### **Question 11 (2 marks)**

**Marks**

Solve  $\log_3 5 = 2\log_3 10 - \log_3 x$ .

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### **Question 12 (2 marks)**

**Marks**

Find the coordinates of the centre and the radius of the circle with equation

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$$x^2 + 8x + y^2 - 6y - 24 = 0$$

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**Question 13 (2 marks)**

A Year 12 class of 24 students were surveyed about the type of exercise they do.

12 go to the gym, 13 run and 4 neither go to the gym nor run.

- (a) Complete the two-way table below to represent this data.

**1**

|            | Gym | Do Not Gym | Total |
|------------|-----|------------|-------|
| Run        |     |            | 13    |
| Do Not Run |     | 4          |       |
| Total      | 12  |            |       |

- (b) Find the probability that a student who runs also goes to the gym.

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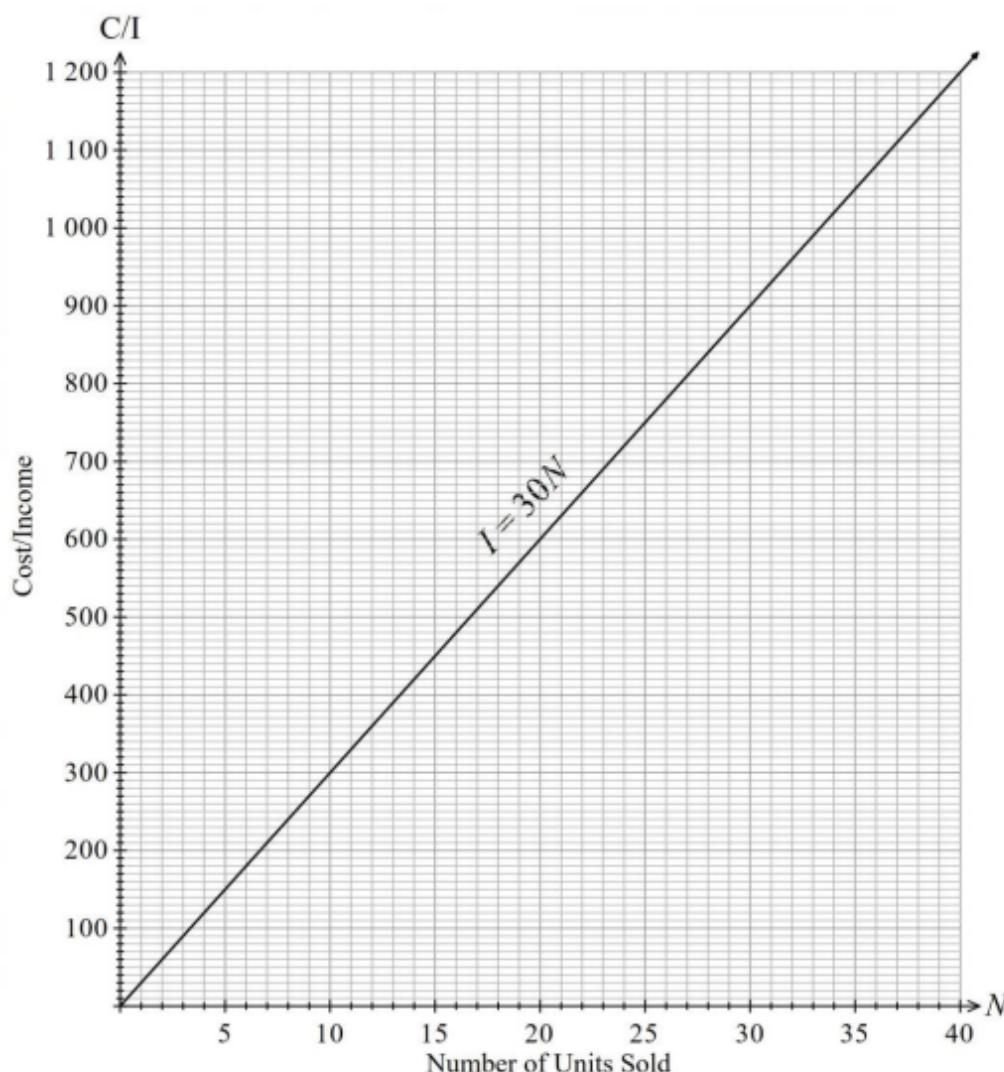
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### Question 14 (5 marks)

Marks

A wholesaler sells a device called a Midien for \$30 each. The income  $I$  from selling  $N$  devices is graphed below.



Each day their fixed costs (wages, rent etc) are \$300 and each Midien costs them \$10.

- (a) The formula for the cost involved in selling  $N$  Midiens in a day is

1

$C = 10N + 300$ . What is the cost when 20 Midiens are sold in a day?

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Question 14 continues on page 13

**Question 14 (continued)**

**Marks**

- (b) Draw the line representing the equation  $C = 10N + 300$  on the graph on page 12.

**2**

- (c) How many Midiens would they need to sell in a day to break even?

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- (d) How much profit or loss would they make on a day where they sold 30 Midiens? Clearly state profit or loss.

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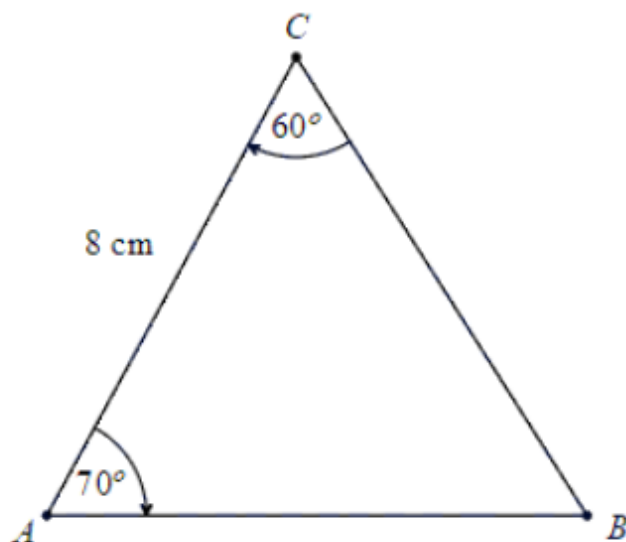
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### Question 15 (3 marks)

Marks

In the diagram below it is given that  $AC = 8\text{ cm}$ ,  $\angle C = 60^\circ$  and  $\angle A = 70^\circ$ .



- (a) Show that the length of  $AB$  is 9.04 cm.

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- (b) Hence, find the area of the triangle to 2 decimal places.

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**Question 16 (3 marks)**

The number of people ( $N$ ) who attend a show at a venue varies inversely with the amount of floor space in  $\text{cm}^2$  allowed per person ( $A$ ). Three hundred people attended the venue when  $1\text{m}^2$  of floor space was allowed per person.

- (a) Find the constant ( $k$ ) of variation.

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- (b) Hence or otherwise, find how many people can attend the venue if each person is allowed  $4500\text{ cm}^2$  of floor space.

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**Question 17 (3 marks)**

Consider the series  $\ln x - 3(\ln x)^2 + 9(\ln x)^3 + \dots$

- (a) Show that these terms form a geometric series. 1

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- (b) Hence, or otherwise, find the values of  $x$  for which the series has a limiting sum. 2

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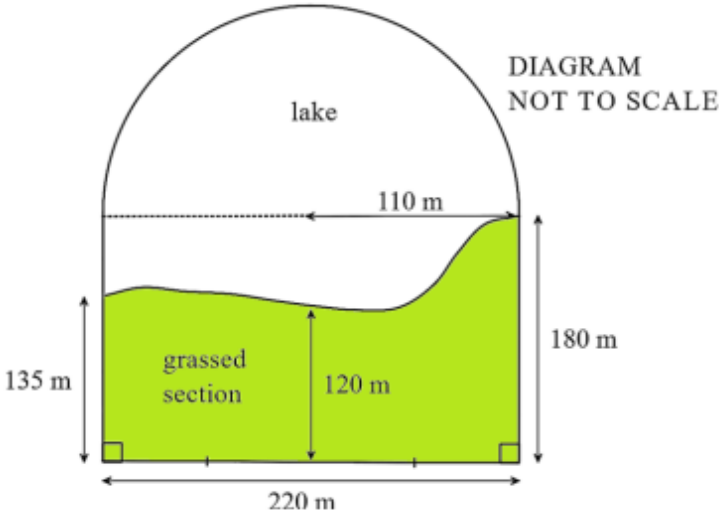
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**Question 18 (3 marks)**

A landscaper, Annabel, wanted to ask the local council to renovate the park near her home. Her suggestion was to have a park partially occupied by a lake and the rest be a grassed section, as shown in the diagram below.



The park consists of a rectangle with dimensions 220 m and 180 m, and a semi-circle with a radius of 110 m. Some measurements from the end of the grassed section to the edge of the lake are also shown.

Using two applications of the Trapezoidal rule, calculate the approximate area of the lake, to the nearest square metre.

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**Question 19 (3 marks)**

The third term in a geometric sequence is 18 and the sixth term is 486.

- (a) Find the first term and the common ratio.

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- (b) Find the sum of first 10 terms of this sequence.

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**Question 20 (3 marks)**

Find the equation of the normal to the curve  $y = 3\log_e 2x$ , when it passes through the point  $P$  whose  $x$  – coordinate is 2. Leave your answer in the form:

$$y - y_1 = m(x - x_1)$$

3

[illegible]

Question 21 (2 marks)

Marks

Given that  $\int_1^6 f(x)dx = 6$  and  $\int_3^6 f(x)dx = -4$ . What is the value of  $\int_1^3 (f(x) + x)dx$  ? 2

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Question 22 (2 marks)

Differentiate  $y = \left( 1 + e^{2x} \right)^8$ . 2

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**Question 23 (5 marks)**

**Marks**

An object moves along the number line so that its displacement ( $x$  metres) from the origin is given by:

$$x(t) = 12 - 2t + 12 \log_e(t + 3)$$

where time  $t$  is measured in seconds.

- (a) Where is the particle initially?

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- (b) Show that the particle is at rest when  $t = 3$ .

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- (c) In which direction is the particle moving when  $t > 3$  ? Justify your statement with mathematical working or in words.

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Question 24 (3 marks)

Marks

- (a) Show that  $\frac{\operatorname{cosec} A}{\operatorname{cosec} A - \sin A} = \sec^2 A.$
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- (b) Hence, or otherwise, find  $\int \tan^2 x \left( \frac{\operatorname{cosec} x}{\operatorname{cosec} x - \sin x} \right) dx.$
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Question 25 (3 marks)

Marks

Evaluate  $\int_1^3 \frac{3x^2 - 1}{5x^3 - 5x + 9} dx$  leaving your answer in simplest exact form. 3

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Question 26 (2 marks)

If  $\sec^2 \theta = 2$  and  $0 \leq \theta \leq \pi$ , find all possible values of  $\theta$ . 2

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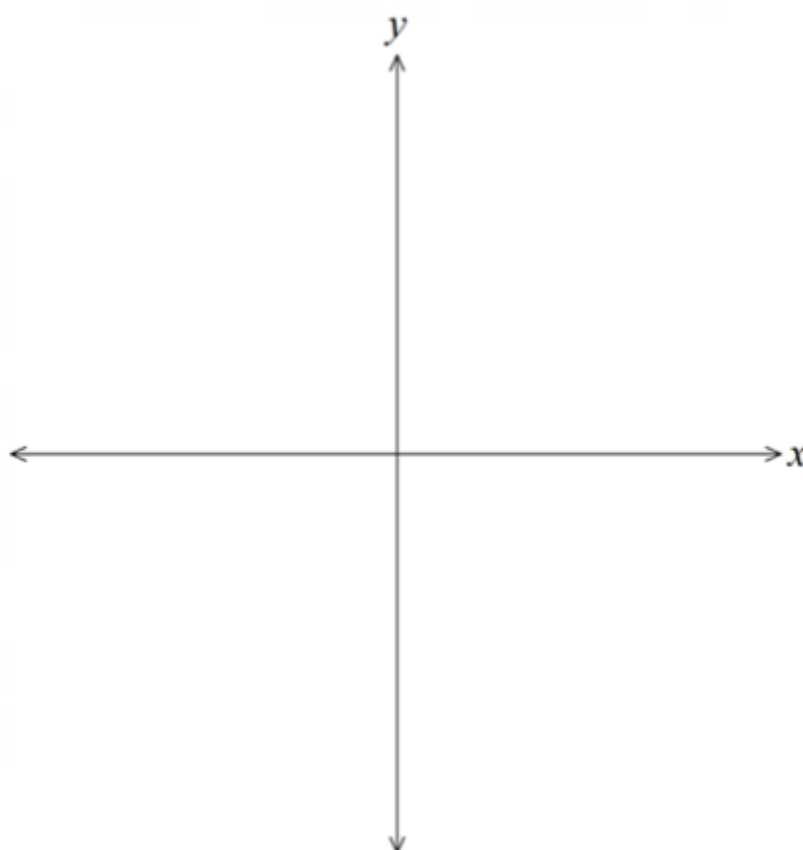
**Question 27 (5 marks)**

**Marks**

(a) Let  $f(x) = (x + 2)^2 - 1$

Plot the graph of  $f(x)$  in the given space below, showing all intercepts and turning point.

**2**



(b)  $f(x)$  is transformed to  $g(x)$  such that,

$$g(x) = 2f(3 - x)$$

Describe the three transformations in the correct order.

**3**

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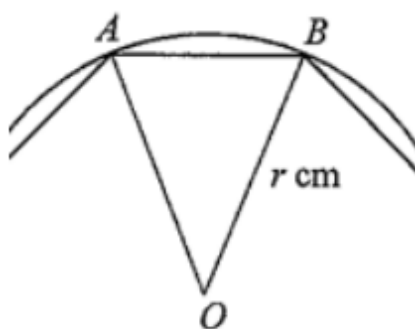
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**Question 28 (4 marks)**

**Marks**

In the diagram  $\triangle AOB$  is a section of a regular ten sided decagon inscribed in a circle with centre  $O$  and radius  $r$  cm.



- (a) Find  $r$  given that the area of the full decagon is  $100 \text{ cm}^2$ .

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- (b) Find the perimeter of the decagon. Give your answer correct to 3 decimal places.

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**END OF SECTION II**

## **Section I - Extra writing space**

**If you use this space, clearly indicate which question you are answering.**

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## **Section I - Extra writing space**

**If you use this space, clearly indicate which question you are answering.**

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# Mathematics Advanced

## Section II Answer Booklet 2

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| Teacher: |
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### Section II

#### Booklet 2 – Attempt Question 29 – 35 (35 marks)

- Instructions**
- Write your Teacher's Name and Student Number at the top of this page.
  - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided on pages 41 to 43 of Booklet 2. If you use this space, clearly indicate which question you are answering.

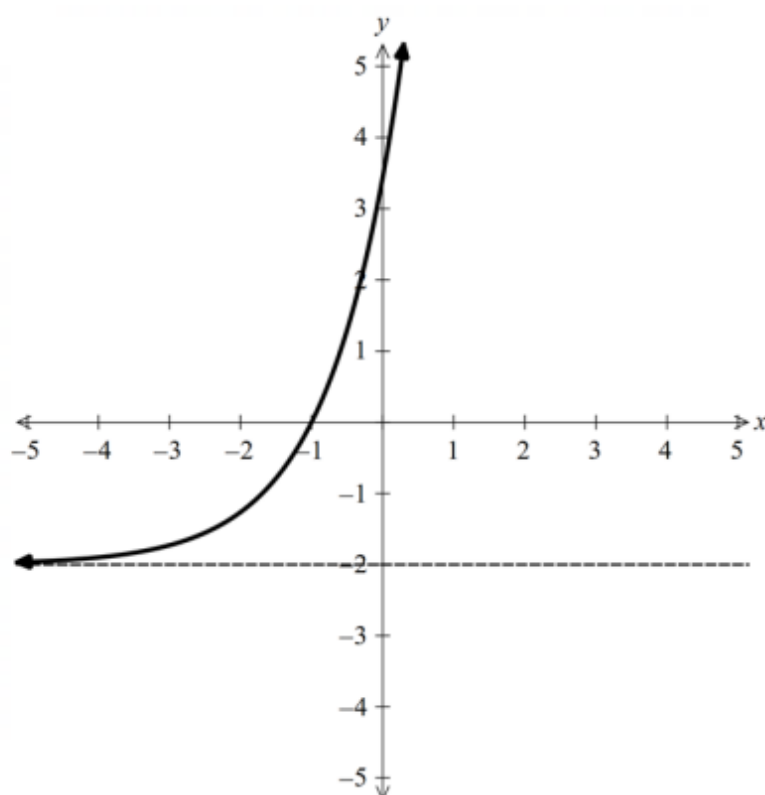
|           |     |
|-----------|-----|
| Q29 – Q32 | /17 |
| Q33 – Q35 | /18 |

**Please turn over**

### Question 29 (4 marks)

Marks

The graph of  $y = 2e^{x+b} + c$  is given below.



- (a) In your own words, justify why  $c = -2$ .

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- (b) Show that  $b = 1$ .

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Question 29 continues on page 31

**Question 29 continued**

- (c) Find the area bounded by the curve, the  $x$  – axis and the  $y$  – axis.

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**Question 30 (3 marks)**

When Lucio plays for his soccer team, they have an 85% chance of winning. When he does not play, their chance of winning is 40%.

The chance that Lucio plays any particular game is 90%.

(a) Find the probability that Lucio plays and his team wins.

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(b) Find the probability that Lucio played, given that his team won.

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**Question 31 (4 marks)**

**Marks**

In a game of chance, a player is to spin a spinner with sectors of unequal sizes.  
The following table gives the probability distribution of the money that the player could win.

|             |               |        |               |        |
|-------------|---------------|--------|---------------|--------|
| Money prize | \$2000        | \$4000 | \$6000        | \$8000 |
| Probability | $\frac{2}{5}$ | $x$    | $\frac{1}{5}$ | $y$    |

The expected value of the game is \$4000.

What is the probability of winning \$6000 or more?

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**Question 32 (6 marks)**

**Marks**

An open topped fish tank of volume  $30 \text{ m}^3$  is to be made in the shape of a rectangular prism with length  $3x$  metres, width  $x$  metres and height  $h$  metres. Materials cost \$20 per square metre for the base of the tank and \$35 per square metre for the sides of the tank.

- (a) Show that the total cost \$C of making the fish tank is given by

**3**

$$C = 60x^2 + \frac{2800}{x}$$

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**Question 32 (continued)**

- (b) Hence find the dimensions of such a tank with the least total cost. Give your answer correct to 3 decimal places.3

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**Question 33 (7 marks)**

**Marks**

Consider the curve  $y = 1 - e^{-x^2}$ .

- (a) Find any stationary points and determine their nature.
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- (b) Show that there are points of inflection at  $x = \pm \frac{1}{\sqrt{2}}$ .
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Question 33 continues on page 37

**Question 33 (continued)**

**Marks**

- (c) What value does  $y$  approach when  $x \rightarrow \infty$ ?

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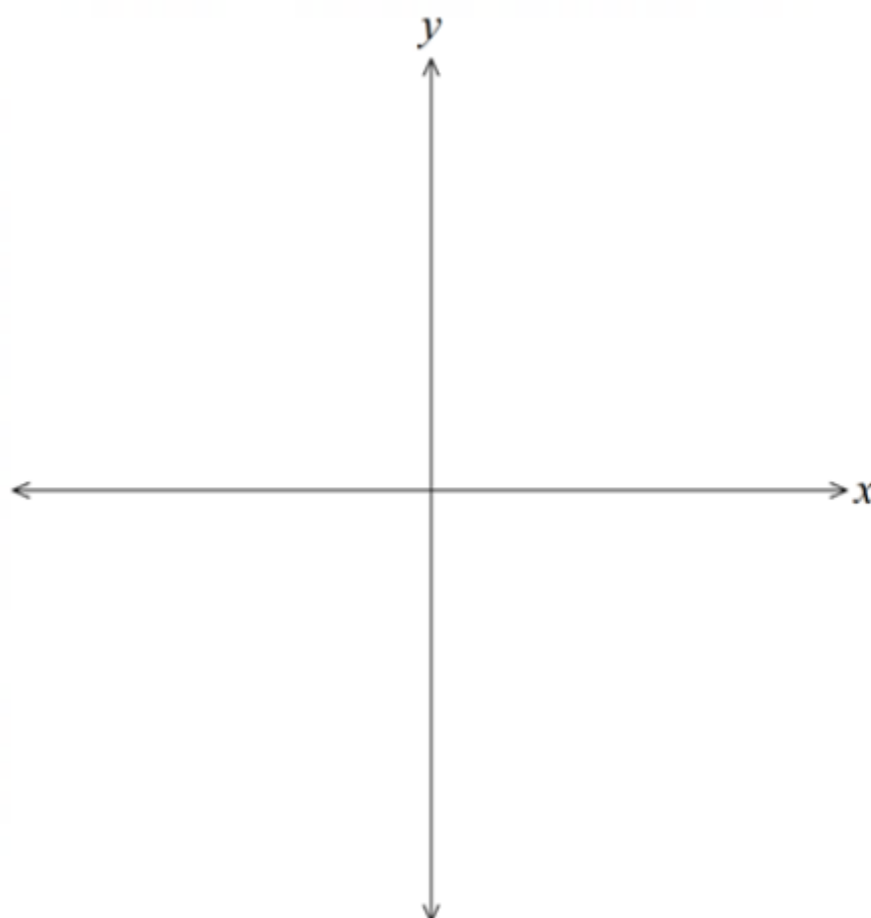
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- (d) Sketch  $y = 1 - e^{-x^2}$ , showing all asymptotes, turning points and points of inflection.

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**Question 34 (7 marks)**

**Marks**

A circular Ferris wheel rotates at a constant rate, and the height of a certain passenger above the ground as a function of time can be modelled by the equation:

$$h(t) = 20 \sin\left(\frac{\pi}{20}(t - 15)\right) + 22,$$

where  $h(t)$  represents the height in metres and  $t$  is the time in seconds.

- (a) What is the maximum height of the person during the ride? **1**

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- (b) Show that the Ferris wheel completes 15 full rotations during a 10-minute ride. **2**

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**Question 34 continues on page 39**

**Question 34 (continued)**

- (c) A passenger can view the harbour from any height above 30m. After the Ferris wheel starts the ride, when will the passenger reach height of 30m for the first time? 2

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- (d) How long will the passenger stay above 30m during the 10-minute ride?  
Answer in minutes and seconds. 2

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**Question 35 (4 marks)**

**Marks**

A substance decomposes at a rate that is proportional to the mass of the substance remaining. The remaining mass ( $M$ ) can be modelled using the formula

$M = M_0 e^{-kt}$ , where  $t$  is the time in hours,  $M_0$  is the initial mass and  $k$  is a constant.

Initially there is 30 kg of the substance. After 24 hours half of the substance has decomposed.

Determine how long it will take for 90% of the substance to decompose.

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**End of Examination**